DESIGN THINKING PROJECT (DTP)

DESIGN ENERGY SYSTEM AND MODELLING UNCERTAINITY REPORT

SOLAR DEHYDRATING SYSTEM

MEMBERS AND CONTRIBUTIONS

STUDENT ID	NAME	CONTRIBUTIONS
1009087	Zhao Hongqing	Built heat exchanger, report
1009171	Ayushi Kalmath	Built heater, report
1009365	Rachel Kok Jing Yi	Built heater, report
1009131	Bay Khean Meng	Built heat exchanger, report
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PROBLEM STATEMENT

How might we reduce food wastage in Singaporean based supermarkets (NTUC, Giant etc..) by reprocessing nearly expired food items through sustainable methods.

SCENARIO

Food waste in Singapore's supermarkets is a persistent issue caused by over-ordering, poor inventory management, and strict cosmetic standards that reject edible but blemished produce. For example, supermarkets like FairPrice only accept completely green bananas, leading to unnecessary disposal when bananas arrive slightly yellow. In 2023, FairPrice alone generated 2,940 tonnes of food waste, despite efforts such as donating $18,000–$24,000 worth of food monthly through programmes like Food from the Heart. A realistic scenario is when a supplier delivers edible bananas that are slightly yellow but are rejected due to appearance standards; with no alternative buyer or storage, the bananas are discarded instead of being redirected to those in need—highlighting the tension between retail practices and food insecurity.

SPECIFICATION

To determine fan specs:

Assuming 60% sunlight absorbed under 0.6m * 0.6m Area.

$$Rate\ of\ heat\ transfer\ from\ sunlight,\ \dot{Q} = Area * \frac{1kW}{m^2} * 60\%$$

$$= (0.6)^2 m^2 * \frac{1kW}{m^2} * 60\% = 216\,W$$

$$Mass\ flow\ needed, \dot{m} = \frac{\dot{Q}}{c\Delta T} = \frac{216W}{1.005\frac{kJ}{kg*K} * 30K} = 0.007164\frac{kg}{s}$$

$$Air\ Flow\ rate, \dot{V} = \frac{\dot{m}}{\rho} = \frac{0.007164\frac{kg}{s}}{1.2\frac{kg}{m^3}} = 5.97 * 10^{-3}\frac{m^3}{s} = 12.65\ cfm$$

Fan specs chosen: $P_{fan} = 15W, V_{fan} = 12V$

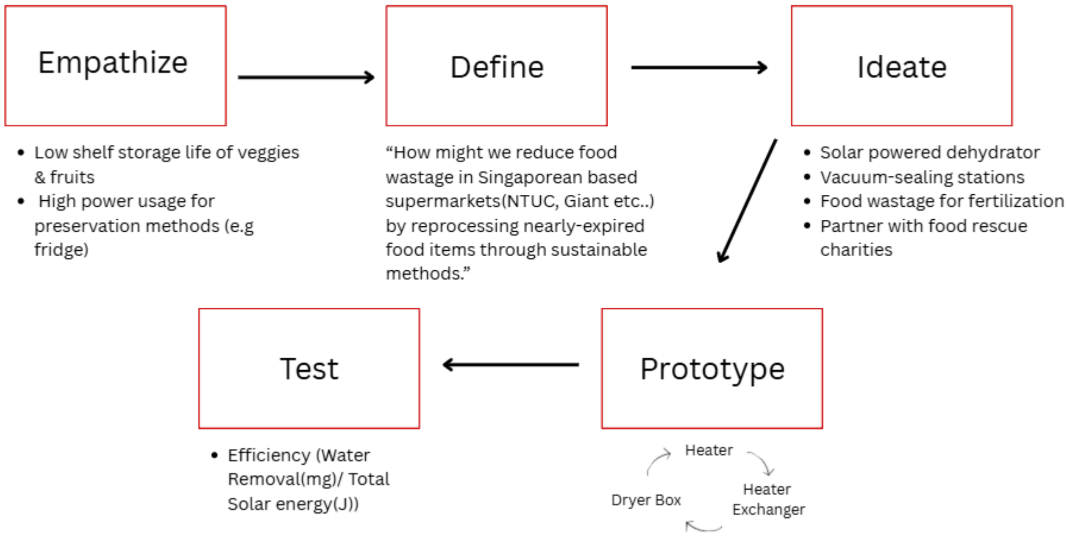
To determine Peak Sun hour needed:

$Daily\ energy\ needed = 15W * 2h = 30Wh$

$Daily\ Load = 1.25A * 2h = 2.5Ah$

$$Peak\ Sun\ Hours\ needed, PSH = \frac{2.5Ah}{70\% * 1.1} = 3.26\ PSH$$

DESIGN PROCESS



SUNLIGHT ANALYSIS OF 5-YEARS:

Goal: Identify the two consecutive months with the lowest average daily Peak Sun Hour(PSH) over five years and, via analysing data **(obtained from the NASA website - appendix(3))** and using confidence intervals and normality checks, confirming it meets the dehydrator's 3.26 h/day requirement; then identify and statistically validate the daily two-hour window of peak irradiance in the month that has the lowest PSH.

Refer to the excel sheet provided in the appendix (4)

Two consecutive months with lowest average daily PSH: **November and December**

Average daily PSH over the two months: **4.3220302 hr/day**

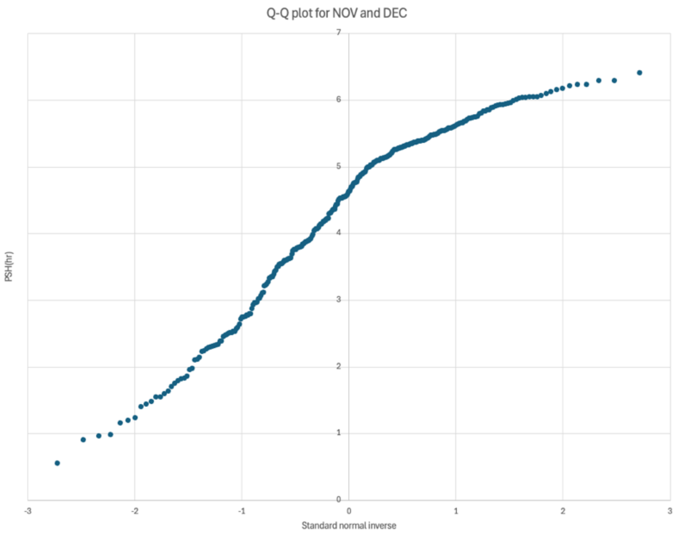


Figure 1: Q-Q plot for the NOV-DEC PSH data

Plotting the Q-Q plot for the Nov-Dec data, we can see that it follows an 'S' shape plot, implying that the distribution is a **normal distribution**, moreover, our dataset has 305 datapoints and the data points are independent and identically distributed, which proves that our data is roughly a **normal distribution using the central limit theorem**

Now we estimate the minimum value our true mean can take by calculating the lower confidence interval. We use T-distribution since we do not know the true variance of our data set.

Calculating the lower bound confidence interval at 99% confidence using: $\left[\bar{x} - t_{(n-1,\alpha)} \frac{S_x}{\sqrt{n}}\ ,\ \infty\right)$

Where, $\bar{x} = 4.3220302$, $n = 305$, $\alpha = 0.01$, $S_x = 1.3494304$ and $t_{(n-1,\alpha)} = 2.3386767$

Substituting the values, we get the interval to be $[4.14132492\ ,\ \infty)$ **which implies that, with a 99% confidence level, the true mean is at least** $4.14132492\ hours/day$ **which is well above the minimum requirement of our prototype(3.26 hours/day).**

Therefore our prototype meets its minimum requirements even in the months with lowest PSH with a 99% confidence level

Month with the least daily average PSH: **December(4.2920123 hr/day)**

The sun faces SE during December, **therefore the ideal direction in which we should place our panel is SE (refer to the link in the appendix(6))**

<u>Refer to the excel sheet provided in the appendix(4)</u>

2 hours of the day during December with the highest average irradiance: **10am-12pm (586.28667 Wh/m^2)**

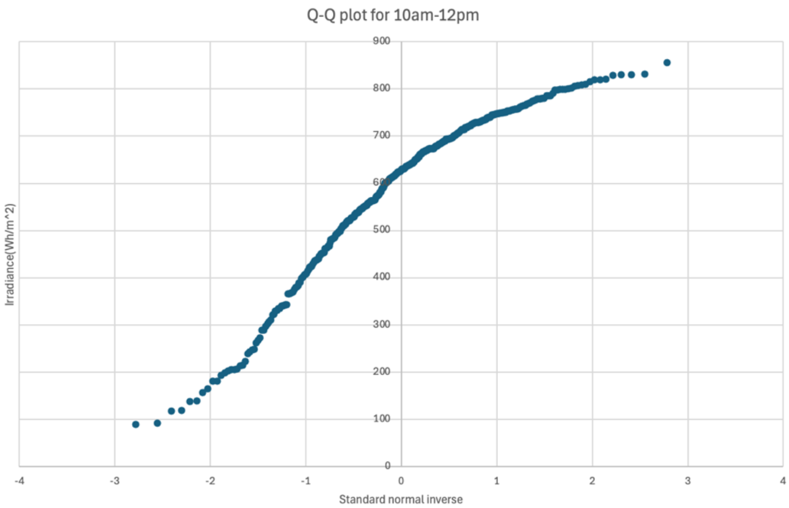


Figure 2: Q-Q plot for 10am -12pm average irradiance data

Plotting the Q-Q plot for the 10am-12pm data, we can see that it follows an 'S' shape plot, implying that the distribution is a **normal distribution**, moreover, our dataset has 372 datapoints and the datapoints are independent and identically, which proves that our data is roughly a **normal distribution using the central limit theorem.**

Now we estimate the minimum value our true mean can take by calculating the lower confidence interval. We use T-distribution since we do not know the true variance of our data set.

Calculating the lower bound confidence interval at 99% confidence using: $\left[\bar{x} - t_{(n-1,\alpha)} \frac{S_x}{\sqrt{n}}\ ,\ \infty\right)$

Where, $\bar{x} = \mathbf{586.28667}$, $n = 372$, $\alpha = \mathbf{0.01}$, $S_x = \mathbf{168.172862}$ and $t_{(n-1,\alpha)} = \mathbf{2.33644097}$

Substituting the values, we get the interval to be $[\mathbf{565.9144044}\ ,\ \infty)$ **which implies that, with a 99% confidence level, the true mean is at least $\mathbf{565.9144044}\ Wh/\mathrm{m}^2$.**

Now we calculate the energy the panel absorbs, assuming that the panels efficiency is 70%:

$Area$ of panel = 0.136 m^2

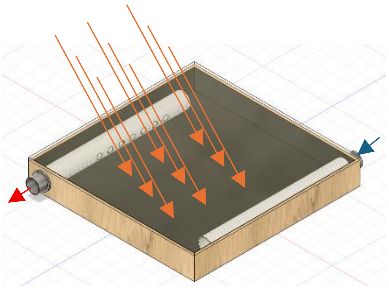
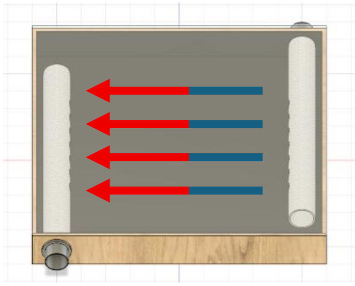
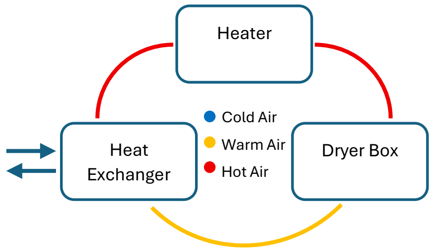
Energy = $Irradiance \times Efficiency\ \times\ Area = 565.9144044 \frac{Wh}{\mathrm{m}^2} \times\ 0.7\ \times\ 0.136\ \mathrm{m}^2 = \mathbf{53.87\ Wh}$

Therefore, **with a 99% confidence level, the true mean yields at least 53.87 Wh of energy which is well above the minimum requirement of our prototype(30 Wh)**

In conclusion, with a 99% confidence level, the optimum time to operate our prototype during the month with the least PSH(December) is from 10AM – 12PM with the panel facing in the SE direction.

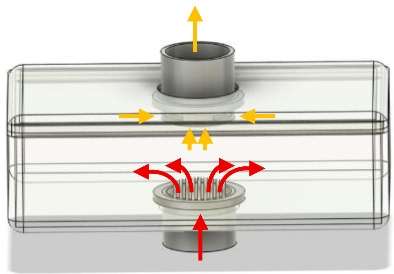
<u>**PROTOTYPE DETAIL**</u>

Heater:

The heater box functions as a medium to collect heat energy from the sun. Its inner walls are painted black to maximize the absorption of sunlight. A clear acrylic sheet covers the box, allowing sunlight to enter while also trapping heat inside through the greenhouse effect. In addition, perforated pipe distributes the air evenly, making sure that the air is evenly heat up before exiting from the heater.

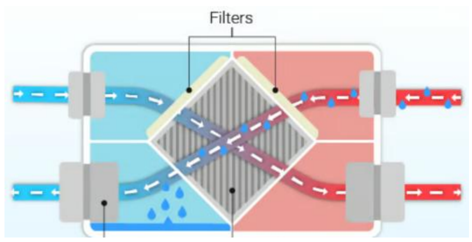
Dryer Box:

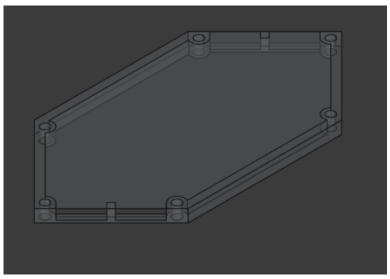
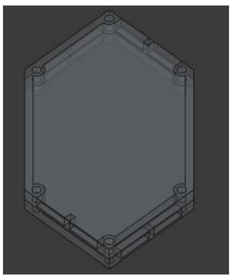


The dryer box serves as the enclosure for the dehydration process. We use a microwave-safe container to withstand the high temperatures generated by the heater. Its inlet and outlet are engineered to optimize airflow: the inlet is a flanged, lock-in cylindrical duct that channels hot air uniformly throughout the container, promoting even heating and drying. The vented-grill outlet then captures and expels humid air from every level, preventing moisture buildup and ensuring efficient dehydration.

Heat Exchanger:

After the hot air leaves the dryer box, the humid air is still hotter than inlet air, so we could recover the heat by using it to heat up the inlet air. This is achieved by a heat exchanger:



The Basic element of it is separator plates and aluminium foil. The 3d printed plates support the foil and forms channels to guide and separate the two streams of air, forming a counterflow heat exchanger. Plates are mound and pressed by bolts and fit inside a wooden box to provide a casing, adapter to piping, as well as sealing. Measured data shows the heat exchanger preheat the inlet air by ~5 degrees, boosting the working temperature of the entire system to ensure better performance.

PROTOTYPE ANALYSIS

(Refer to Appendix (7))

Initial weight of sponge + water before drying = 197.5 g (**refer to appendix(5, figure 4))**

Final weight of sponge + water after drying = 115.2 g (**refer to appendix(5, figure 5))**

Surface area of the dehydrator = $0.6\ \text{m} \times 0.6\ \text{m} = 0.36\ \text{m}^2$

Surface area of solar panel = $0.136\ \text{m}^2$

Time taken to dry the sponge (t) = 1.5 hours (from 11:20AM to 12:50PM)

Average Solar Intensity (I) during the time interval = 834.2727273 W/m²

$$\boldsymbol{Solar\ Energy}_{\text{absorbed}} = I \times A \times t = \left(834.2727273\ \frac{\text{W}}{\text{m}^2}\right) \times (0.36\ \text{m}^2 + 0.136\ \text{m}^2) \times (1.5\ \text{h})$$

$$= 620.6989\ \text{Wh}$$

$$\boldsymbol{Efficiency} = \eta = \frac{\text{mass of water evaporated (g)}}{\text{total solar energy absorbed (Wh)}} = \frac{197.5\ \text{g} - 115.2\ \text{g}}{620.6989\ \text{Wh}} = \mathbf{0.1332\ g/Wh}$$

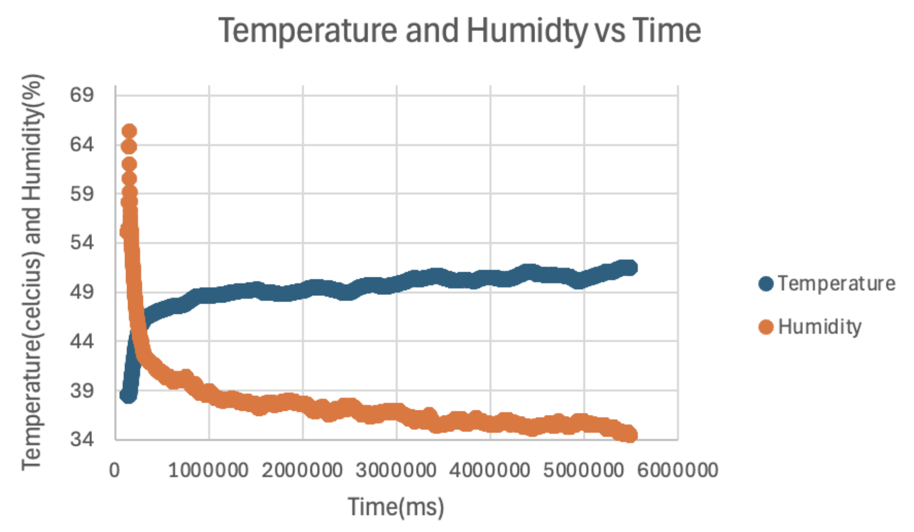


Figure 3 3:Temperature and Humidity vs Time graph of our prototype

Figure 3 shows exactly the conditions you want for food drying:

- **Warm, steady heat:** The box quickly climbs to around 48–52 °C and then holds there for the full 1.5 hours —right in the sweet spot for removing moisture without cooking the food.
- **Dropping humidity:** Humidity plummets from over 60 % down to the mid-30 % range (**ideal percentage for drying fruits**), meaning the air inside continuously pulls water out of the sponge (or food) rather than saturating.

Together, a **stable 50 °C** plus **low, falling humidity** creates a consistent driving force for moisture removal—making it ideal for dehydrating fruits, vegetables, or meats.

COST ANALYSIS:

S.NO	PART OF THE PROTOTYPE	ITEM	Area	Cost
1.	**Heater**	Plywood (Bottom)	59cm X 60cm	$8.88
		Plywood (sides)	60*7.5 X4(sides)	$4.52
		Acrylic	59 cm X 60cm	$10.99
		PVC pipe	Length-47cm X2 Diameter-5cm	$4.7
		Black plastic paper	----	$0.18
2.	**Dryer box**	Plastic box	----	$5.43
3.	**Heat exchanger**	Aluminium foil	Area of 6 hexagon aluminium sheet-931.6cm^2	$1.52
		Plywood (top and bottom)	20cm X 11cm X 2	$0.94
		Plywood (sides)	20cm X 7cm X2	$0.6
		Plywood(sides)	11cm X 7cm –4X (area of circles) =11cm X 7cmX2 - $4X(\pi \times (2.25)^2)$	$0.19
4.	**Other components**	PVC pipe	Length-22.5cm+24cm+25cm+11.5cm+11cm Diameter-5cm	$4.70
		Plywood	Support- area-1252cm^2	$2.64
		Blower Fan	---	$8
		20W Solar panel	---	---
			Total:	**$53.29**

At a build cost of $53.29 per unit, our solar dehydration system delivers professional-grade performance at roughly one third the price of commercial system. By making it a budget-friendly device, more businesses, small-scale producers and researchers get access to this sustainable and eco-friendly drying technology. Ultimately, it will significantly reduce food waste.

CONCLUSION

This project presents a sustainable and scalable solution to reduce food wastage in Singaporean supermarkets by reprocessing nearly expired food through controlled dehydration. The prototype consisting of a **heater**, **heat exchanger**, and **enclosed dryer box** is solar-powered and designed for on-site integration within supermarket settings to support food preservation.

Performance testing demonstrated that the system maintained a consistent drying temperature of **48–52 °C** and effectively reduced internal humidity levels from over **60% to the mid-30% range (which is ideal for drying fruits)**, creating optimal conditions for moisture removal without compromising food integrity. Using a sponge simulation to represent high-moisture food, the system achieved an efficiency of **0.1332 g/Wh**, highlighting its capability to convert solar energy into useful thermal energy for dehydration.

Constructed using **readily available** and **reusable** materials, the prototype is energy efficient, cost effective, and well-suited for **commercial use** in Singaporean supermarkets supporting both sustainability goals and food waste reduction.

APPENDIX

1. https://www.ntuc.org.sg/uportal/news/FairPrice-Combats-Food-Waste

2. https://www.straitstimes.com/singapore/unsold-but-not-unwanted

3. https://power.larc.nasa.gov/data-access-viewer/

4. https://sutdapac-my.sharepoint.com/:x:/g/personal/bharath_rajeev_mymail_sutd_edu_sg/ERl5wus_AjJBma3G5crGSK0BZnaV2AQcujcyZAYiKrtJDQ?e=OWNpny

5. 

Figure 4: Weight of sponge before drying.

Figure 5: Weight of sponge after drying

6. https://www.suncalc.org/

7. https://sutdapac-my.sharepoint.com/:x:/g/personal/kheanmeng_bay_mymail_sutd_edu_sg/ETE7qMa46gtOrsPxHqK_drYBoEQIOUgL28hy5dox-5Quxw?e=UKutSm